YEAR 10 PHYSICS, AQA GCSE, LESSON 4 OF 8

PARTICLE MODEL OF MATTER

LESSON 04

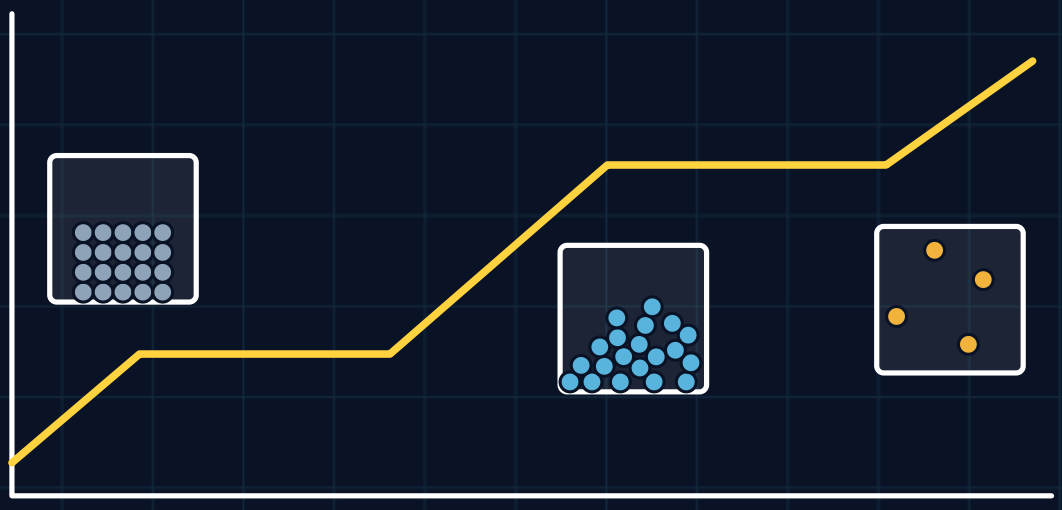
MEASURING DENSITY (REQUIRED PRACTICAL)

A jeweller is offered a lump of “gold” with an awkward shape. How could its density be measured?

IN THIS LESSON YOU WILL

- measure the density of a regular solid from its dimensions
- measure the volume of an irregular solid by displacement
- measure the density of a liquid
- choose instruments and explain how to make the measurements accurate

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 4** of the Particle Model of Matter unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW

Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT

The explanation and diagrams. Fill in each example from the slides with your teacher, then try "Your turn" alone. Read this again when you revise.

TRY IT

Practice that builds up: first with support, then on your own, then a challenge. Every second lesson ends with the Quick check from the slides.

PRACTICAL

Record your measurements and work out results from them.

EXAM CORNER

AQA-style questions with a tip on how the marks are given.

REVISE IT

Summary, cover-and-answer questions and an "I can" check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are **close together** **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01	Solids, liquids and gases	02	Gas particles and pressure
03	Density	**04**	**Measuring density (required practical)** Regular solids, irregular solids by displacement, liquids; resolution and sources of error (slides 18–23)
05	Internal energy and heating curves	06	Specific latent heat
07	Specific heat capacity	08	Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-**

model

Lesson

04

MEASURING DENSITY (REQUIRED PRACTICAL)

AQA Physics 4.3.1.1, required practical 5; Combined Science 6.3.1.1, required practical 17

BIG QUESTION **A jeweller is offered a lump of "gold" with an awkward shape. How could its density be measured?**

BY THE END YOU CAN

- measure the density of a regular solid from its dimensions
- measure the volume of an irregular solid by displacement
- measure the density of a liquid
- choose instruments and explain how to make the measurements accurate

displacement | **measuring cylinder** | **meniscus** | **resolution** | **vernier callipers** | **balance**

YOU WILL NEED

Per group: balance (0.1 g or better), 30 cm ruler, vernier callipers, regular blocks, irregular objects (stones, modelling-clay lumps) on thread, 100 cm³ measuring cylinder, displacement can, beaker, water, a second liquid (salt water or cooking oil), paper towels.

Teaching deck, slides 18–23
panphy.app/gcsephy/decks/particle-model/#18

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Write the equation for density.

$\rho = \frac{m}{V}$

2
How many cubic centimetres are there in 1 millilitre?

1 ml = 1 cm³

3
How do you calculate the volume of a cuboid?

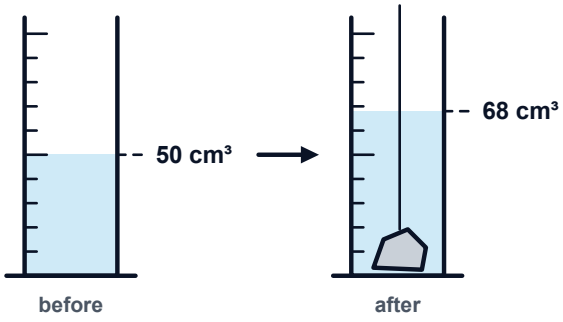
volume = length × width × height

EXPLAIN

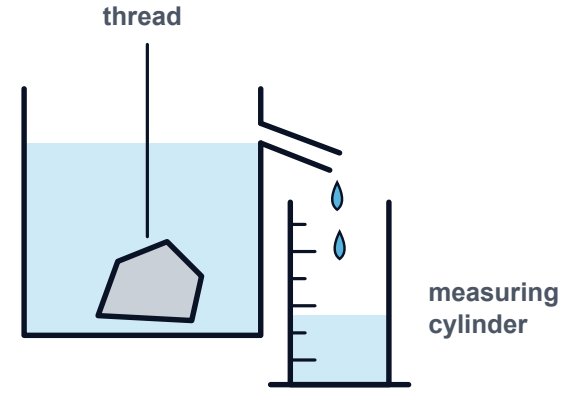
TWO MEASUREMENTS, ONE CALCULATION

To find a density you always need two measurements: the **mass**, from a balance, and the **volume**. Then $\rho = \frac{m}{V}$. The mass is easy. The method for the volume depends on what you are measuring.

	HOW TO FIND THE VOLUME
Regular solid (cube, cuboid)	Measure its length, width and height with a ruler or vernier callipers. Volume = length × width × height.
Irregular solid (stone)	**Displacement.** Lower it into water. The volume of water it pushes out of the way is equal to its own volume.
Liquid	Pour it into a measuring cylinder and read the scale.



Volume of stone = 68 − 50 = 18 cm³



displacement can, filled to the spout

For an object too big for the cylinder

MAKING IT ACCURATE

- Read a measuring cylinder at **eye level**, at the **bottom of the meniscus** (the curved surface of the liquid).
- **Vernier callipers** read to 0.1 mm or better; a ruler reads to 1 mm. The instrument with the smaller scale division has the better **resolution**.
- Measure the mass **before** the object gets wet: water on it would add to the mass.
- Check that the balance reads **zero** before you start.
- Lower the object gently on a thread so that no water splashes out, and tap away any air bubbles stuck to it.

> **Get the marks:** The width of a measuring cylinder matters. A narrow cylinder has scale lines further apart for each cm³, so you can read it more precisely. Use the narrowest one the object will fit into.

INVESTIGATE

REQUIRED PRACTICAL: DENSITY

Wipe up spills at once. Carry glassware with two hands.

STAY SAFE

- Water on the floor is a slip hazard: mop up spills straight away.
- Do not drop metal blocks: keep them in the middle of the bench.
- Report any broken glass to your teacher. Do not pick it up.

☐ PART A: REGULAR SOLIDS

1. Measure the length, width and height of each block.
2. Calculate its volume.
3. Measure its mass on the balance.
4. Calculate its density.

1 Record your results.

Material	Length / cm	Width / cm	Height / cm	Volume / cm³	Mass / g	Density / g/cm³
aluminium	4.0	3.0	2.0	24	64.8	2.7
steel	4.0	2.0	2.0	16	125.6	7.9
wood (oak)	5.0	4.0	2.0	40	28.0	0.70

☐ PART B: IRREGULAR SOLIDS

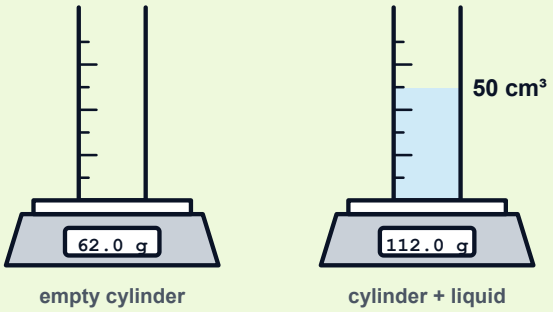
1. Measure the mass of the dry object.
2. Half fill a measuring cylinder with water. Record the volume.
3. Lower the object in on a thread until it is completely under water. Record the new volume.
4. Volume of object = second volume − first volume. Calculate the density.

2 Record your results.

Object	Mass / g	Volume of water / cm³	Volume of water and object / cm³	Volume of object / cm³	Density / g/cm³
stone	46.8	50	68	18	2.6
clay lump	27.0	50	65	15	1.8

PART C: A LIQUID

1. Measure the mass of the empty measuring cylinder.
2. Pour in about 50 cm³ of the liquid. Record its exact volume.
3. Measure the mass of the cylinder and liquid.
4. Mass of liquid = second mass − first mass. Calculate the density.



empty cylinder | cylinder + liquid

3 Record your results.

Liquid	Mass of empty cylinder / g	Mass of cylinder and liquid / g	Mass of liquid / g	Volume of liquid / cm³	Density / g/cm³
water	62.0	112.0	50.0	50	1.00
salt water	62.0	117.5	55.5	50	1.11

PRACTISE

TRY IT

4 Densities in g/cm³: oak 0.70, aluminium 2.7, iron and steel 7.9, copper 8.9. Suggest what each of your regular blocks is made of. Do your densities match exactly? Suggest one reason why not.

Sample: 2.7 g/cm³ is aluminium and 7.9 g/cm³ is iron (steel). Small differences come from measuring the sides only to the nearest millimetre, or from blocks that are not perfect cuboids.

5 Draw a ring around the correct answer in each box.

A measuring cylinder should be read at

the top of the meniscus	**the bottom of the meniscus**	the edge of the glass

Compared with a ruler, vernier callipers have

a better resolution	a worse resolution	the same resolution

6 A student measures the volume of a stone by displacement. An air bubble is stuck to the stone. Explain how this affects the density that the student calculates.

The bubble pushes extra water out of the way, so the measured volume is too big.

Density = mass ÷ volume, so the calculated density is too small.

7 **CHALLENGE** A cork floats, so it cannot be pushed under water by its own weight. Suggest how its volume could still be found by displacement.

Push it just under the surface with a thin pin (or tie it to a small metal sinker whose volume has been measured, and subtract the sinker's volume).

CHECKPOINT

QUICK CHECK

Slide 23, Lessons 3–4. Answer each question, then check it as your teacher reveals the answer.

1
A block has a mass of 150 g and a volume of 60 cm³. Calculate its density.

$\rho = \frac{150}{60} = 2.5$ g/cm³

2
Write 2.5 g/cm³ in kg/m³.

2500 kg/m³

3
A liquid has a density of 800 kg/m³. Calculate the mass of 0.50 m³ of it.

$m = 800 \times 0.50 = 400$ kg

4
Why is a gas much less dense than a liquid?

Its particles are far apart, so there is less mass in each cm³.

5
The water level rises from 40 cm³ to 55 cm³. What is the volume of the object?

15 cm³

6
Where do you read the scale on a measuring cylinder?

At the bottom of the meniscus, at eye level.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: A 6-mark method is marked on **whether someone could follow it**. Name each instrument, say what it measures, and finish with the calculation.

8 A student has a small, irregularly shaped stone. Describe a method the student could use to determine the density of the stone. **[6 marks]**

Indicative content: measure the mass of the dry stone with a balance; part fill a measuring cylinder with water and record the volume; lower the stone in on a thread until fully submerged; record the new volume; volume of stone = difference between the two readings (or use a displacement can and collect the water); read the scale at eye level, at the bottom of the meniscus; density = mass ÷ volume; repeat and calculate a mean. Level 3 (5–6): a detailed, logically ordered method that would give a valid result.

9 A metal cube has sides of 2.0 cm and a mass of 71.2 g. Calculate the density of the metal in g/cm³. **[3 marks]**

VOLUME	$V = 2.0 \times 2.0 \times 2.0 = 8.0$ cm³ (1)
EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{71.2}{8.0}$ (1)
ANSWER	8.9 g/cm³ (1) unit:

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Density needs a mass (balance) and a volume.
- Regular solid: volume = length × width × height.
- Irregular solid: volume = volume of water displaced.
- Liquid: volume from a measuring cylinder; mass = mass of cylinder and liquid − mass of empty cylinder.
- Read the bottom of the meniscus at eye level.

COVER THE PAGE AND ANSWER

1 Name the method for finding the volume of an irregular object.

2 Where on the meniscus do you read a measuring cylinder?

3 How do you find the mass of a liquid in a measuring cylinder?

4 Which has the better resolution: a ruler or vernier callipers?

I CAN…

- measure the density of a regular solid
- use displacement to find the volume of an irregular solid
- measure the density of a liquid
- explain how to make the measurements accurate
- describe the method in a 6-mark answer

REFERENCE

EQUATION TOOLKIT

Every equation from Lessons 1–4. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find V from $\rho = \frac{m}{V}$, cover V. m is above ρ, so $V = \frac{m}{\rho}$.

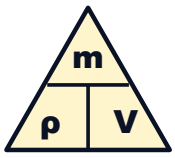
$$\rho = \frac{m}{V}$$

density = mass ÷ volume

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$

Lesson 3



1
Write what you know. Convert first: g → kg, cm³ → m³, kJ → J.

2
Write the equation in symbols.

3
Rearrange if you need to, then substitute the numbers.

4
Give the answer with a unit. Ask: is it sensible?

REFERENCE

KEY WORDS FROM THIS LESSON

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Displacement: Finding the volume of an object from the volume of water it pushes out of the way

Meniscus: The curved surface of a liquid in a tube; read the bottom of it

Resolution: The smallest change in a quantity that an instrument can show

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. $\rho = \frac{m}{V}$
2. 1 ml = 1 cm³
3. volume = length × width × height

COVER THE PAGE

1. Displacement.
2. At the bottom, at eye level.
3. Subtract the mass of the empty cylinder from the mass of the cylinder with the liquid.
4. Vernier callipers.